

Investigation of BOLD signal intensities in long COVID patients using 7T functional MRI

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ABSTRACT

Long COVID is increasingly associated with disruption in brain homeostasis, manifesting as severe neurological dysfunction, brain fog and cognitive impairment. This present study investigated localised cognitive deficits in long COVID patients by examining brain blood oxygenation-level-dependent (BOLD) signal activity using ultra-high-field 7 T (7T) task-based functional magnetic resonance imaging (fMRI). Whole-brain BOLD signal differences were assessed across 19 long COVID patients, and 27 healthy controls (HC) including 12 COVID-recovered (Cov-RHC) and 15 COVID-19-naïve HC (nHC). 225 fMRI volumes were acquired during the Stroop colour-word task. Functional and anatomical images were processed using SPM12 to extract the BOLD signal intensity time course from whole-brain voxels for inferences between cohorts during task-fMRI. Significantly low BOLD activation in long COVID patients was observed compared to Cov-RHC in the anterior cingulate cortex ($p = 0.002$, cluster size=650, Z -value = 4.67), and the precuneus ($p = <0.001$, cluster size = 1893, Z -value = 4.67). Furthermore, BOLD intensities in precuneus showed a negative association with self-reported pain scores ($p = 0.040$) and the duration of illness ($p = 0.03$) in long COVID patients, suggesting significant correlation between BOLD signal and an increase in duration of illness and pain levels. No statistically significant BOLD differences were observed for inter-group comparisons between nHC vs. long COVID, and nHC vs. Cov-RHC. Response times to incongruent ($p = 0.002$) and congruent task stimuli ($p = 0.001$) significantly varied between nHC and long COVID cohorts, demonstrating overall faster information processing by nHC. Reduced BOLD signals to 'core' brain regions in long COVID imply reduced cognitive control by intrinsic networks that mediate information processing, cognitive and executive functions due to perturbations linked to cerebral blood flow, oxygenation status, and ongoing neuroinflammation.

1. Introduction

Post-acute sequelae of SARS-CoV-2 infection or long COVID is defined as the development of new symptoms three months or symptom persistence for more than four weeks following SARS-CoV-2 infection (CDC, 2024; Raveendran, 2021; Soriano et al., 2020). Neurocognitive deficits, particularly cognitive dysfunction, memory and attention impairments are frequently reported symptoms, coupled with physical and mental fatigue in long COVID (Crivelli et al., 2022; Hartung et al., 2022; Z. Li et al., 2023; Lopez-Leon et al., 2021; Mazza et al., 2021; Premraj et al., 2022; Velichkovsky et al., 2023). These cognitive complaints may be attributed to persistent neurobiological changes, including widespread central and peripheral neuroinflammation, vascular

disruption and microstructural brain changes (Colombo et al., 2021; Lee et al., 2021; Lindan et al., 2021). The exact neuroinvasive mechanisms in long COVID, however, remain poorly understood (Tai et al., 2023).

Growing neuroimaging evidence in long COVID indicates alterations across cortical and subcortical regions (Ajčević et al., 2023; Dacosta-Aguayo et al., 2024; Dadsena et al., 2025; Du et al., 2023; Qin et al., 2021) with grey matter and white matter heterogeneity (Bendella et al., 2023; Besteher et al., 2022; Douaud et al., 2022; Hafiz et al., 2022; Latini et al., 2022; Paolini et al., 2023; Thapaliya et al., 2023). These changes are consistent with cognitive impairment, suggesting large-scale network dysfunction related to cognitive complaints in long COVID (Dacosta-Aguayo et al., 2024). Functional magnetic resonance imaging (fMRI) uses blood oxygenation level-dependent (BOLD) to

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detect brain activity patterns (Ogawa et al., 1992) and report functional alterations in hippocampus (Saikarthik et al., 2025), frontal and temporal regions (Li et al., 2023). Widespread connectivity changes are observed within sensory, motor, attention, salience and default mode networks (Baraniuk et al., 2024; Churchill et al., 2023; Dacosta-Aguayo et al., 2024; Esposito et al., 2022; Mohammadi and Ghaderi, 2024; Muccioli et al., 2023; Voruz et al., 2023).

More recently, a multimodal study in long COVID cohorts yielded significant differences in functional connectivity while no comparable differences were detected in structural and perfusion analyses when compared to COVID-recovered individuals (Fineschi et al., 2024). These findings suggest that post-COVID-19 cognitive dysfunction may arise from dysregulated attention networks. A critical gap, therefore, exists in understanding executive processes, particularly within brain regions that support task-based cognitive control and executive functions in long COVID, healthy and COVID-recovered individuals.

While the literature base in long COVID research is rapidly expanding (Baraniuk et al., 2024; Diez-Cirarda et al., 2024; Graf et al., 2025; Inderyas et al., 2026; Langan et al., 2022; Liou et al., 2024; Rane Levendovszky et al., 2025; Rua et al., 2024; Talkington et al., 2024; Thapaliya et al., 2023, 2025a, 2025b, 2025c; Trufanov et al., 2025), the application of 7 T (7T) fMRI remains considerably underexplored that offers unprecedented advantage over conventional methods in permitting detection of subtle neurovascular and functional dynamics (Kreitz et al., 2023). The Stroop colour-word task, being a well-validated paradigm probes for selective attention, response inhibition and cognitive interference resolution (Russo et al., 2023), which are core executive functions frequently reported in long COVID (Baraniuk et al., 2024).

Therefore, the goal of this study was to localise neural dysfunction by eliciting robust BOLD responses to the Stroop task in regions critically implicated in cognitive control. We reasoned that long COVID patients would exhibit reduced signal intensity during cognitive Stroop task performance, particularly in brain regions governing conflict monitoring, working memory and sustained attention. The present study assessed these functions across three participant groups: long COVID patients (n = 19), COVID-recovered healthy controls (n = 12), and COVID-naïve healthy controls (n = 15) using 7T fMRI by detecting subtle haemodynamic changes during cognitively demanding task performance. We hypothesised that: (1) significant task-evoked BOLD differences would be observed during cognitively demanding incongruent trials in brain regions corresponding to executive and cognitive functions; (2) prolonged response times to incongruent trials would reflect slowed cognitive processing speed; and (3) BOLD signal intensity would correlate with patient-reported clinical scores.

2. Materials and methods

2.1. Participant recruitment

This cross-sectional study was approved by the Griffith University Human Research Ethics Committee (2022/666) with written informed consent obtained from all participants. The present study was conducted at the National Centre of Neuroimmunology and Emerging Diseases (NCNED) on the Gold Coast, Queensland, Australia. Nineteen long COVID patients fulfilling the WHO criteria for post-COVID-19 by Delphi consensus (Soriano et al., 2022) were recruited to undergo MRI scans on a 7T scanner. Twenty-seven healthy controls (HC) aged between 18 and 65 years with no reported chronic health condition or evidence of underlying illness were also included in the study. Individuals with chronic psychological illness, malignancies, autoimmune, neurological, cardiovascular diseases, pregnancy and/or breastfeeding were excluded. All participants were requested to complete online research registry questionnaires developed by NCNED. These questionnaires incorporate the CDC chronic fatigue syndrome inventory and a series of validated quality of life (QoL) measures that could be accessed via Research Electronic Data Capture (REDCap). The questionnaire specifically

sought information on previous history of COVID-19 infection, or multiple infections and approximate dates of infection(s) and whether the diagnoses were made through COVID-19 polymerase chain reaction (PCR test) or a Rapid Antigen Test. Out of 27 HCs included in the study, 15 individuals self-reported having no history of COVID-19 infection 12 participants reported a complete recovery following COVID-19 infection and were healthy with no underlying illness or comorbidities.

The clinical measures were obtained from all the participants, similar to our previous publications (Inderyas et al., 2026; Thapaliya et al., 2025a). Quality of life (QoL) measures were acquired using 36-Item Short Form Survey (SF-36) (Alonso et al., 1995). SF-36 scores ranged between values from 0 to 100, and an average was assigned to calculate for each domain. Duration of illness was estimated from the age of symptom onset for long COVID patients, and severity of ‘pain’ was assessed on a scale of 0 to 100 points, where values closer to 0 indicated extremely severe pain symptoms (higher levels of pain) and values closer to 100 presented no symptoms (higher QoL), (see Table 1). Detailed demographic information and QoL measures for duration of illnesses, pain, general and mental health scores and physical function measures for participants are provided in the Supplementary file.

2.2. MRI acquisition

MRI data were acquired using the acquisition protocols described in our previous publication (Inderyas et al., 2026). 19 long COVID patients and 27 HCs were scanned on a 7T whole-body MRI scanner (Siemens Healthcare, Erlangen, Germany) with a 32-channel head coil (Nova MRI Wilmington NC, USA). 225 functional volumes were acquired sagittally using a multiband echo-planar imaging (EPI) pulse sequence developed at the University of Minnesota (Auerbach et al., 2013) while participants responded to the Stroop colour-word task. The acquisition parameters were: repetition time (TR) = 2000 ms, echo time (TE) = 22.4 ms, flip angle = 70°, multi-slice mode = Interleaved, acquisition matrix 192x192 and voxel size = 1.25 mm³. 60 Stroop stimuli were presented during the task fMRI that lasted for 7.5 min. Anatomical sagittal T1-weighted data was acquired using a Magnetization Prepared 2 Rapid Acquisition Gradient Echo (MP2RAGE) Sequence with TR = 4300 ms, TE = 2.45 ms, first inversion time (TI1) = 840 ms, second inversion time (TI2) = 2370 ms, first flip angle (FA1) = 5°, second flip angle (FA2) = 6°, and resolution = 0.75 mm³ with matrix size = 256x300 x320.

2.3. The Stroop colour-word task experiment

All participants performed a cognitively challenging Stroop colour-word paradigm during fMRI. For the task trial, two coloured words were displayed with the upper word as either ‘Blue’, ‘Red’, ‘Yellow’,

Table 1
Demographic information for 19 long COVID patients and 27 healthy controls, sub-grouped into Cov-RHC and nHC, number of females and males in each group, age, sex, clinical measures for pain and illness duration with mean and standard deviations (±). (Also see Supplementary Tables for symptom scores).

	Long COVID (n = 19)	HC (n = 27)	
		Cov-RHC (n = 12/ 27)	nHC (n = 15/ 27)
Age	47.77 ± 12.7	34.4 ± 6.3	37.5 ± 11.3
Two-tailed P-value for age	0.002 ^a 0.0008 ^b 0.40 ^c		
Sex (f/m)	14/5	8/4	10/5
Duration of illness	0.78 ± 0.67	n/a	n/a
Pain scores	49.26 ± 21.12	87.5 ± 17.76	91.3 ± 15.6

n/a denotes not applicable for healthy controls. a = p-value (19 long COVID versus 27 HC), b = p-value (19 long COVID versus 12 Cov-RHC), c = p-value (12 Cov-RHC versus 15nHC).

'Green' or 'XXXX' and coloured in red, green, blue or yellow on a black background. The lower word 'Blue', 'Red', 'Yellow' or 'Green' was in white displayed against black background. Participants responded to whether the colour of the upper word agreed with the meaning of the lower word and decided with 'yes' or 'no' by pressing one of the two buttons on the handpiece provided during the scan. As shown in Fig. 1, neutral stimulus contains upper word as 'XXXX', incongruent with upper word written in a different colour from its meaning, 3) and congruent with upper word written in the colour of its meaning. The idle period, called "rest", is the pause between a trial response and the next trial onset during which a stationary cross appears on the screen for a randomised period. The task lasted for 7.5 min.

Conflict (interference) occurs with the upper word because of the automatic response to read the upper word to determine its colour instead of the meaning of the lower word. This dual-choice for the congruent-incongruent was used to test information processing ability, cognitive control and executive function in participants by manipulating the difficulty level. The response selection conflict using selective attention would also force the participant to take time to ponder, gaining enough time for inhibitory cognitive interference from conflicting word-link presentations to kick in (Peskar et al., 2023). 40% of randomised trials were incongruent, 30% congruent, and 30% neutral, and the average inter-stimulus time was 10.5 s.

The stimulus-onset and reaction times to the stimulus were recorded for each trial type, and their difference was categorised as response time (RT). Response accuracy was extracted by calculating accuracy from the correct responses over total responses (accuracy = correct/total responses) for each trial type for each group. Two-sample t-tests were performed to estimate significance between cohorts. SPSS was used to perform independent sample t-tests and to estimate effect size to quantify group differences in RT during incongruent and congruent tasks.

2.4. MR image processing

Structural and functional MRI data were visually inspected for quality control before pre-processing with SPM12 running in MATLAB 2019b (The MathWorks, Inc.). SPM works on the principle of using raw images to estimate regionally specific effects through statistical parametric maps (Flandin and Novak, 2020). MRI data were pre-processed using SPM12 for realignment and unwarping, slice time correction, and co-registration to anatomical images (Friston et al., 1994). Resampling was performed to an isotropic 1 mm, followed by spatial normalisation into MNI space. Smoothing was performed at 5 mm FWHM. Quality control measures were performed at each preprocessing step.

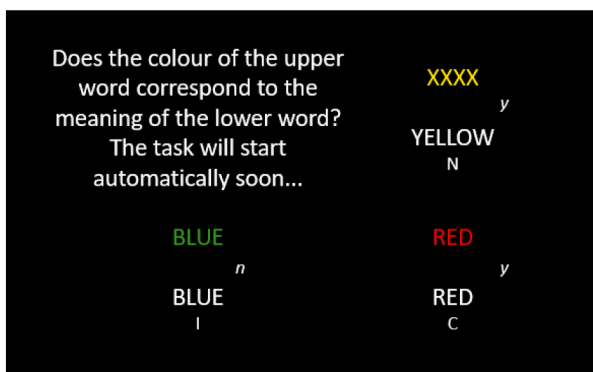


Fig. 1. Illustration of the Stroop colour-word task with four examples displayed on the screen to the participants. Top left: instructions, top right: neutral (N); lower left: incongruent (I); lower right: congruent (C); on the right y indicates answer yes, n indicates answer no to instructions. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Time-series inspection of signal dropouts was assessed for each participant. An a-priori threshold of 1.5 mm was established for framewise displacement (FD) for exclusion criteria. All study participants maintained an FD below 1.5 mm indicating good data quality across the three cohorts. Fig. 2 provides detailed preprocessing steps of MRI data.

2.5. General Linear Modelling (GLM) and Statistical Modelling

A GLM design matrix was used for a mass-univariate approach to analyse the BOLD data based on the experimental and confounding effects. Residual variability was then used to derive univariate statistics at each voxel to compute statistical parametric maps. 5 initial BOLD volumes were removed manually, and the remaining 220 volumes were used for 'first-level' fMRI models for within-subject analysis. "Incongruent" and "Congruent" conditions were defined by their onset and duration periods in the experimental design matrix. FD and realignment parameters, along with white matter and CSF time-series were used as regressors to limit confounding effects. GLM applied these statistical parametric models at each voxel and the 'Estimates' of the GLM parameters were explored by using contrast vectors to produce statistical parametric maps. These estimates were used to generalise the results to the population in 'second-level analysis' where it calculates the standard error, isolating the variability in parameter estimates and performing T-tests for the mean for the contrast to be tested for significance. Multiple group comparison analyses were performed across subjects. BOLD differences were measured for: 19 long COVID vs 15 nHC, 19 long COVID vs 12 Cov-RHC, and 15 nHC vs 12 Cov-RHC.

Default settings were used for 'apply mask', and 'p-value adjustment to control' option 'none', for the whole-brain analysis. We used the voxel "threshold T or p-value" of 0.001 to locate clusters that were composed of individual voxels passing that set threshold. Cluster size threshold was set to 10 to require at least 10 adjacent significant voxels. Statistical inference was measured with the false discovery rate corrected cluster p-value (cluster p_{fdr}). BOLD time-series values from the two clusters were extracted using MARSBAR in SPM12 and a Pearson test was performed to estimate the strength of the correlations between BOLD time-series values and clinical measures acquired from long COVID patients. Age and gender were used as control variables to account for confounding effects and adjust for differences in our analysis.

3. Results

3.1. Long COVID vs Cov-RHC

Group comparison between long COVID patients and Cov-RHC showed significant BOLD differences for several voxel clusters. Refining the extent threshold to 600 for limiting active voxels displayed, we obtained two clusters with significant BOLD signals. We found significantly reduced BOLD activation in long COVID patients within the anterior cingulate cortex (ACC, Fig. 3) ($p_{fdr} = 0.002$, cluster size = 650; coordinates = -5,47,-6; Z-score = 4.67); and precuneus ($p_{fdr} < 0.001$, cluster size $kE = 1893$; coordinates = 4,-59,17; Z-score = 4.67) (see Fig. 3).

We did not find any significant differences in the BOLD signal between 19 long COVID vs. 15 nHC, and 15 nHC vs. 12 Cov-RHC.

3.2. Correlations with clinical measures in Long-COVID

A significant negative correlation was observed between BOLD signal in the precuneus and pain levels ($p = 0.040$, $r = -0.553$, $CI [-0.81, -0.98]$); and duration of illness ($p = 0.03$, $r = -0.526$, $CI [-0.79, -0.09]$); see Fig. 4.

We did not find any significant relationship between ACC BOLD signal intensity and pain ($p = 0.258$, $r = -0.324$); or duration of illness ($p = 0.171$, $r = -0.348$). No significant correlations existed for the BOLD values from the two cluster regions and general health, mental health, or physical function scores.

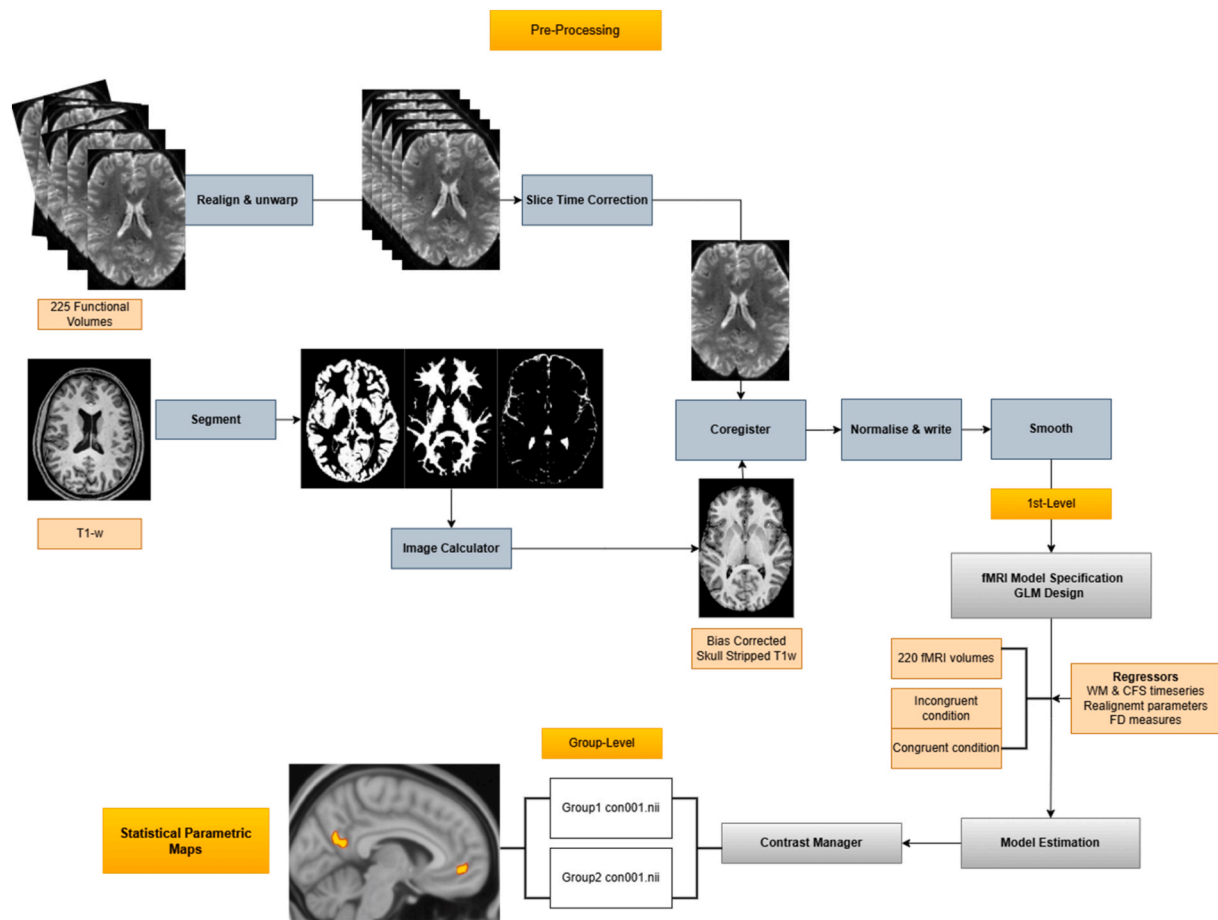


Fig. 2. Illustration of the preprocessing steps for analysing BOLD time-series between 19 long-COVID and 27 healthy cohorts. Following spatial and temporal realignment of the 225 fMRI volumes, the realigned and slice time corrected. fMRI volumes were registered to the bias-corrected skull-stripped T1w, co-registered and normalised to standard MNI space. Smoothing was performed using a 5 mm Gaussian kernel. Within-subject 1st-level analysis was performed by omitting the first five volumes for magnetization correction. fMRI model was specified using onset and duration times of the incongruent and congruent conditions; white matter and cerebrospinal fluid time-series, realignment measures from the first step 'Realign & unwarp', and framewise displacement (FD) measures as regressors. Between-group 2nd-level analysis used contrasts files from the 1st-level analysis, sorting subjects into two groups. Finally, statistical parametric maps (SPM) of significant BOLD activity differences were obtained using age and gender as nuisance regressors. (See supplementary file for more details).

3.3. Accuracy and Response times to the Stroop Tasks

We tested for differences between RT and accuracies of those responses to the incongruent and congruent stimuli of the Stroop paradigm. Overall, no statistically significant differences were observed in task accuracies across all cohorts. RT significance across cohorts was tested and Hedges g was then estimated for RTs using independent samples effect sizes as accurate representation of small sample size.

- nHC and long COVID patients presented significant differences in their RTs while performing both incongruent ($p = < 0.05$, Hedges $g = -1.08$) and congruent ($p = 0.001$, Hedges $g = -1.16$) tasks. Long COVID patients showed delayed responses to the tasks (incongruent RT; $mean \pm SD = 2.053 \pm 0.44$) and (congruent RT; $mean \pm SD = 1.79 \pm 0.46$).
- However, no significant differences were detected for RT between long COVID and Cov-RHC groups while performing incongruent ($p = 0.29$, Hedges $g = -0.42$) and congruent ($p = 0.1$, Hedges $g = -0.53$) tasks.
- RT differences between nHC and Cov-RHCs were also not statistically significant for incongruent ($p = 0.2$, Hedges $g = -0.47$) and congruent ($p = 0.1$, Hedges $g = 0.58$).

- Overall RT differences were significant between long COVID patients and all 27 HC during incongruent ($p = 0.01$) and congruent ($p = 0.006$) conditions, (see Table 2).

4. Discussion

To the best of our knowledge, this is the first study to document BOLD signal differences in long COVID and COVID-recovered healthy controls (Cov-RHC) using 7 T Stroop task fMRI. We report significantly lower BOLD signal activation in 19 long COVID patients, compared to 12 Cov-RHC within brain regions of the anterior cingulate cortex (ACC) and precuneus. Relative to Cov-RHC, long COVID patients showed reduced BOLD activity in response to cognitively challenging incongruent vs. congruent Stroop colour-word task stimuli. The cingulate cortex is widely associated with emotion and limbic pathways, and is considered integral in response selection, motor and self-control, visuospatial processing, memory and problem-solving (Paxinos and Mai, 2004; Vogt and Gabriel, 1993). Its anterior section, the ACC, corresponds to vital Brodmann regions, and is a key contributor in error and conflict detection, task initiation, maintenance and task feedback (Dosenbach et al., 2007; Paxinos and Mai, 2004).

Likewise, the precuneus is an important node of the default mode network (DMN) and has extensive cortical and subcortical connections to widespread areas, including the ACC and the posterior cingulate

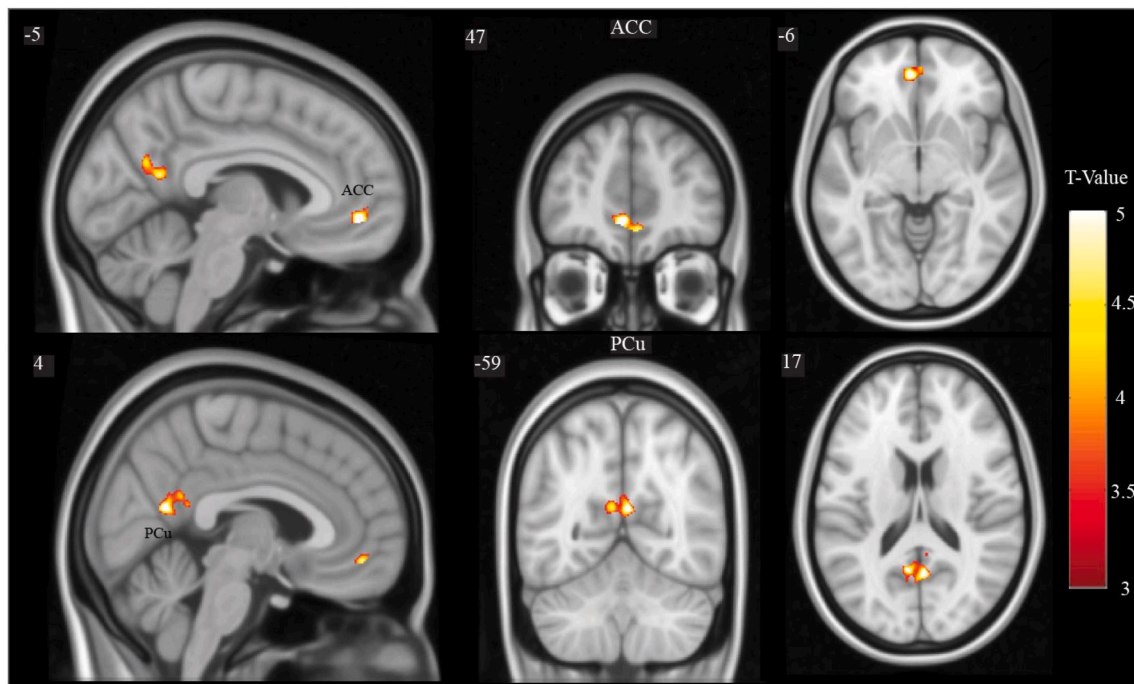


Fig. 3. Voxel clusters with reduced BOLD activation in the ACC (top row) and precuneus (PCu) (bottom row) of long COVID patients compared to COVID-recovered healthy controls, sagittal view shows voxel activation in both regions.

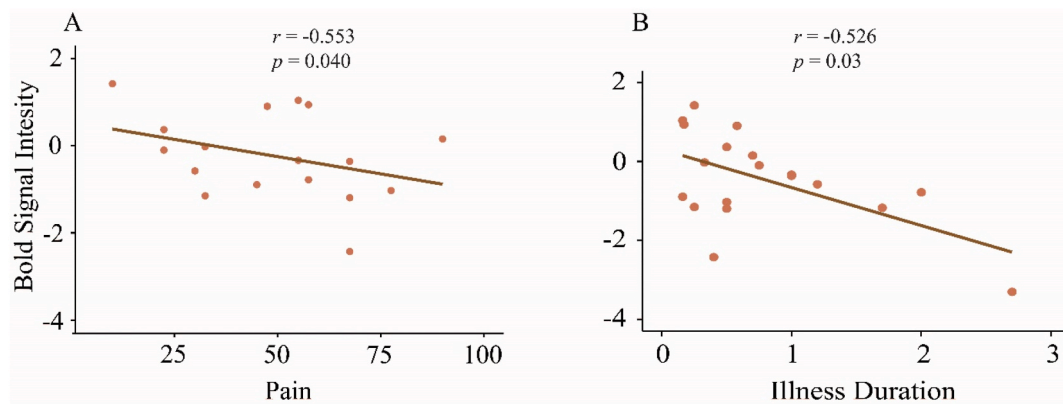


Fig. 4. BOLD signals correlations with Pain and Illness Duration. A) Correlation graph between BOLD time-series from precuneus and pain measures in 19 long COVID patients. The x-axis shows pain scores ranging from 0 to 100, where values closer to 0 indicate high pain levels, and values close to 100 signify low pain levels. The y-axis shows the range of BOLD signal intensity values. B). Correlation graph between BOLD time-series values from precuneus and duration of illness in 19 long COVID patients. The x-axis shows the duration of illness in years; the y-axis shows the range of BOLD signal intensity values.

cortex (Fransson and Marrelec, 2008). Collectively with other DMN hubs, the precuneus and posterior cingulate cortex are believed to serve as convergence nodes for information processing (Fransson and Marrelec, 2008; Vincent et al., 2006) and are involved in self-reflection, higher-order neurocognitive processes, motor coordination and attention (Cavanna and Trimble, 2006). Subdivisions of the precuneus show activation during higher order complex cognition (Lyu et al., 2021; Margulies et al., 2009). Crucially, the ACC is central to higher-level cognitive control for monitoring informational conflicts (Botvinick et al., 2004). It is postulated that the ACC identifies conflict between competing conditions such as incongruent Stroop stimuli (with different word and colour) and recruits areas including the dorsolateral prefrontal cortex (DLPFC), both belonging to the central executive network. Therefore, greater conflict in information would imply greater ACC signal thereby facilitating higher cognitive control, increased attention and subsequently higher task performance (Carter and van

Veen, 2007). Concurrently, the DLPFC is also functionally connected to precuneus subdivisions for executive functions and these functions are found to be impaired in the presence of focal precuneus lesions (Yeager et al., 2022). Widespread regional disturbances in the amplitude of low-frequency fluctuation (ALFF) are being reported (Jin et al., 2024; R. Li et al., 2023) implicating precuneus and other task regions as potential sites of post-COVID brain dysfunction.

Based on the evidence, it is plausible that in long COVID, ACC and precuneus fail to amplify the BOLD signal sufficiently (thus the observed low BOLD activation) to recruit the DLPFC for effective conflict resolution during task performance. Therefore, we can reasonably state that cognitive and executive functioning are compromised in long COVID patients due to low BOLD activity within these regions. These subtle alterations may also be reflected at the level of connectivity, suggesting increased susceptibility to mental overload and impaired cognition (Fineschi et al., 2024; Inderyas et al., 2026).

Table 2

Lists the average response accuracy and the response times (RT) in seconds, directed to perform incongruent and congruent Stroop task stimuli. Standard deviations ($\pm$) and p-values are provided, with significant p-values observed for 19 long COVID versus 27 HC, and 15 nHC vs 19 long COVID.

	Long COVID (<i>n</i> = 19)	All HC (<i>n</i> = 27)	Cov-RHC (<i>n</i> = 12)	nHC (<i>n</i> = 15)	p-value
Incongruent Accuracy	0.93 $\pm$ 0.05	0.93 $\pm$ 0.1	0.93 $\pm$ 0.1	0.93 $\pm$ 0.1	0.95 ^a 0.99 ^b 0.95 ^c 0.93 ^d
Congruent Accuracy	0.972 $\pm$ 0.05	0.98 $\pm$ 0.02	0.98 $\pm$ 0.02	0.98 $\pm$ 0.02	0.37 ^a 0.34 ^b 0.6 ^c 0.5 ^d
Incongruent RT	2.053 $\pm$ 0.44	1.68 $\pm$ 0.5	1.825 $\pm$ 0.6	1.5 $\pm$ 0.3	0.01 ^a 0.29 ^b 0.2 ^c <0.05 (0.002) ^d
Congruent RT	1.79 $\pm$ 0.46	1.41 $\pm$ 0.3	1.53 $\pm$ 0.4	1.31 $\pm$ 0.2	0.006 ^a 0.14 ^b 0.15 ^c <0.05 (0.0010) ^d

(mean $\pm$ SD), *a* = *p*-value (19 long COVID versus 27 HC), *b* = *p*-value (19 long COVID versus 12 Cov-RHC), *c* = *p*-value (12 Cov-RHC versus 15nHC), *d* = *p*-value (15nHC versus 19long COVID).

This is consistent with the delayed response times observed during cognitively demanding incongruent Stroop task performance in the long COVID patient group, suggestive of impaired communications between networks. Our recent comparative study likewise documented similar observations of delayed response times in long COVID and chronic fatigue syndrome patients, linking exertional fatigue and network dysfunction with sluggish performance (Baraniuk et al., 2024). Other studies also report attentional lapses and slower processing speed to cognitive exertions in post-viral fatigue syndromes (Baraniuk et al., 2024; Baraniuk et al., 2024; Guo et al., 2021; Orтели et al., 2022), associating cognitive difficulties with fatigue and symptom severity.

While we expected to see significant BOLD signal differences between COVID-naïve normal HC and long COVID patients (19 long COVID vs 15 nHC), no significant clusters survived the activation cluster threshold. This is an interesting observation that needs to be tested for statistical rigour in larger cohort size. Nonetheless, current results emphasise the complex nature of post-viral fatigue syndromes, including CFS, accounting for brain differences that are so obscure, subtle and heterogeneous in nature to suggest distinct brain compensatory reorganisation following recovery.

Alternatively, minimal task-relevant circuit recruitment evident from sluggish RTs in long COVID could reflect similar biological dynamics between long COVID and recovered individuals, unlike those with no history of COVID-19 infection. A more plausible interpretation of the BOLD differences observed in long COVID and Cov-RHC can be deduced from observations that persistent immune activation leads to potential endothelial dysfunction and vascular leakage, with recovered individuals still reporting cognitive deficits in the absence of active COVID-19 symptoms (Hampshire et al., 2021; Petersen et al., 2023). Widespread fMRI ALFF in COVID-19 survivors suggests COVID-recovered healthy individuals do not achieve 100% recovery (Du et al., 2022). Such subtle differences observed as subtle connectivity differences in Cov-RHC, demonstrating longer RT (1.825 $\pm$ 0.6) than normal HC (1.68 $\pm$ 0.5) RT in the present study, are consistent with earlier evidence from the literature.

It may be warranted to question the effect of age on the observed BOLD differences between long COVID and Cov-RHC. Although normal ageing and neurodegeneration may lead to altered BOLD signals, it is essential to consider that BOLD signals changes in normal ageing can also involve adaptive mechanisms (Lee and Kim, 2022). Moreover, haemodynamic equivalence, a non-tenable concept in BOLD fMRI studies, generally assumes that all components of the neurovascular coupling (NVC) system are age-equivalent when generalised across groups (West et al., 2019). Yet diseased adults often show the presence

of glial and vascular changes distinct from healthy older adults. Mean age of our long COVID cohort (~47years) places our sample in the young/middle-age group, outside a geriatric age group. Consequently, neural factors still dominate the BOLD signals differences observed. Furthermore, implication of brain regions in our results is not isolated observation but is supported by emerging evidence of changes within these areas (Kandemirli et al., 2020; Douaud et al., 2022; Braga et al., 2023; Li et al., 2023; Petersen et al., 2023; Madden et al., 2025; Nelson et al., 2025). Above all, null findings in 19 long COVID vs 15 nHC strongly suggest that the observed BOLD signal differences against Cov-RHC are neurological consequences of SARS-CoV-2 exposure and not due to haemodynamic aging artifact.

4.1. Correlation of BOLD signal with Pain and Duration of illness

BOLD signal intensity in precuneus showed a strong negative correlation with the clinical measures of pain and duration of illness in long COVID patients. ~20.7% of variation in BOLD signal within the precuneus ($R^2 = 0.207$) can be explained by the pain levels whereas an even stronger negative correlation was observed for the illness duration suggesting almost 32% of changes in BOLD intensity ($R^2 = 0.318$) in the precuneus may be derived from disease progression. Overall, as pain levels and illness length increased, a decline in BOLD activity was observed in long COVID patients. Pain and post-exertional malaise are common presentations in long COVID.

Although no clear mechanisms have been documented, prolonged viral exposure and inflammatory responses have been suggested as the prime mechanism (Khoja et al., 2022) underlying these BOLD signal deficits. Neuroinflammation, either due to direct (Muri et al., 2023; Natoli et al., 2020; Umesh et al., 2022) or indirect viral invasion, contributes to altered vascular dynamics and neural activity, ultimately leading to a reduction in neuronal conduction (Bispo et al., 2022; Sasso et al., 2022). Viral-driven fusion of neurons and glial cells precipitates neuronal malfunctioning due to imbalances in neurochemical phenotypes (Thapaliya et al., 2025b) and demyelination (Khodanovich et al., 2022). Altered metabolism in the precuneus and cingulate cortex is frequently observed to have neuropsychological consequences (Boeve, 2012; Coutinho et al., 2015; Hugon et al., 2022; Lu et al., 2020). Structural alterations highlight grey matter volume reduction within the frontal and cingulate regions (Lu et al., 2020; Sanabria-Diaz et al., 2022) and reduced cortical thickness in areas including the precuneus (Cataldo et al., 2024) are long-term consequences of COVID-19 infection. Nociceptive pain with prolonged peripheral neuropathy is associated with central nervous system-driven symptoms which correlate with the

length of the disease from the prolonged systemic inflammatory response (Calvache-Mateo et al., 2023; Fernández-de-Las-Peñas et al., 2021). Continuous and chronic disruptions of neurovascular units leading to an altered neurovascular coupling response (Owens et al., 2024) ultimately contribute to the altered BOLD signals, thereby reducing overall cognitive function in patients (Cataldo et al., 2024; Mohammadi and Ghaderi, 2024).

Nonetheless, our study provides preliminary evidence of the altered BOLD signals in patients suffering from long COVID. Weak haemodynamic coupling in regions of the anterior cingulate cortex and precuneus has important long-term cognitive implications in chronically suffering long COVID individuals relative to COVID-recovered HC. Adaptive or maladaptive alterations in the morphology and functionality of neurons involving complex underlying molecular cascades may underlie these changes (Ben Ezzdine et al., 2025). Additionally, adaptive mechanisms can compensate for deficits within the highlighted regions in the COVID-recovered population, which may be similar to normal healthy participants who have never had SARS-CoV-2 infection (Jordan, 2021).

BOLD activation in long COVID also correlated to pain and the duration of illness of the disease while behavioural measures of cognitive conflict during task performance were inferred from response times, which provided a more objective estimate of the consequent cognitive and executive deficits in long COVID patients. The use of 7T fMRI allowed us to detect more subtle modulations and small activation clusters that contribute to the ongoing manifestations of long COVID symptomatology. Consequently, cognitive processing is altered, which can be flagged through BOLD signal changes. These effects may also extend to the periphery, evolving into peripheral neuropathy and manifesting as the chronic pain and fatigue observed in long COVID. Findings from this study contribute to the growing literature of neurocognitive deficits linked to ACC and precuneus and offer a potential basis for further research in BOLD signal investigation in long COVID.

5. Limitations

This study has some limitations. Although the effects of age and sex were accounted for during the analysis, a larger sample size with age and sex matched participants would enhance rigour in future studies for greater insight into the functional changes reported in this present study between COVID-recovered and people who have never had COVID-19. Age acts as a confounding factor and may possibly influence minor perturbations in the neural-vascular coupling system to induce age-related HRF alterations. Likewise, a small sample size could lead to sampling bias and sample-specific findings that impede reproducibility. Inclusion of longitudinal data is required for a deeper understanding of the evolution and resolution of brain changes over time. Patients self-reported COVID-19 infection following rapid antigen testing and were more than 15 days post infection with no further serology testing undertaken.

Although long COVID patients received a formal diagnosis based on the WHO criteria, the evolving nature of this case definition served as the basis of cohort selection in this study. Future analyses may aim at modelling response delays to clarify whether slow behavioural responses correspond to delayed or absent neural activations. Nevertheless, findings reported in this research provide preliminary groundwork for further investigation to probe functional changes in long COVID patients, COVID-recovered and people who have never had COVID-19.

6. Conclusions

Neurocognitive consequences manifested as altered BOLD activation intensities during cognitive task performance in patients diagnosed with long COVID. Findings revealed significant BOLD activity within the anterior cingulate cortex and the precuneus cortex, potentially indicating underlying tissue damage and impaired regional metabolism. This was further reflected by behavioural evidence of delayed response

times to the cognitive task, corroborating the functional relevance of the observed neural alterations. Subsequent clinical correlation analysis identified a negative association between BOLD signal intensity within the precuneus, pain severity and duration of illness, suggesting these may be contributing factors to reduced neurological functioning in the region. However, chronic inflammation and the subtle nature of these neurological alterations warrant further systematic investigation.

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CRediT authorship contribution statement

Maira Inderyas: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft. **Kiran Thapaliya:** Supervision, Validation, Writing – review & editing. **Sonya Marshall-Gradisnik:** Funding acquisition, Supervision, Visualization, Writing – review & editing. **Leighton Barnden:** Funding acquisition, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no competing interests and that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. We have nothing to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bbih.2026.101266>.

Data availability

Data will be made available on request.

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